

This is the title

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Abstract

Keywords: keyword1, keyword2, keyword3

In this paper there are examples of all the useful things you can do with $\text{\LaTeX}$. This document includes settings and examples for using the following packages: amsthm, appendix, array, booktabs, cancel, ccicons, comment, empheq, endnotes, enumitem, float, graphicx, indentfirst, lastpage, lipsum, listings, makeidx, mathtools, multicol, multirow, palatino, paracol, setspace, standalone, tabulararray, tabularx, verbatim, wrapfig, hyperref, geometry, xolor, tcolorbox, textpos, tikz, tkz-euclide, titlesec, eucal, mathdesign, inconsolata.

1 Introduction

This paper is a template for writing $\text{\LaTeX}$ papers for a variety of audiences.

In this paper there are examples of all the useful things you can do with $\text{\LaTeX}$. This document includes settings and examples for using the following packages: amsthm, appendix, array, booktabs,

cancel, ccicons, comment, empheq, endnotes, enumitem, float, graphicx, indentfirst, lastpage, lipsum, listings, makeidx, mathtools, multicol, multirow, palatino, paracol, setspace, standalone, tabulararray, tabularx, verbatim, wrapfig, hyperref, geometry, xolor, tcolorbox, textpos, tikz, tkz-euclide, titlesec, eucal, mathdesign, inconsolata.

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*The author wishes to thank poop.

†The authors also wish to thank their families.

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1.1 The quick brown fox

The quick brown fox jumps over the lazy dog. Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

2 This Section

In section 2 there are examples of the various types of lists¹:

Item 1 header This is item one...The quick brown fox jumps over the lazy dog.

Item 2 header This is item one...The quick brown fox jumps over the lazy dog.

Item 3 header This is item one...The quick

brown fox jumps over the lazy dog.

- 1) **Item 1 header** — This is item one...The quick brown fox jumps over the lazy dog.
- 2) **Item 2 header** — This is item two...The quick brown fox jumps over the lazy dog.
- 3) **Item 3 header** — This is item three...The quick brown fox jumps over the lazy dog.

- **Item 1 header** — This is item one...The quick brown fox jumps over the lazy dog.
- **Item 2 header** — This is item two...The quick brown fox jumps over the lazy dog.
- **Item 3 header** — This is item three...The quick brown fox jumps over the lazy dog.

3 The Other Section

In section 3 there are examples of mathematics:

$$y = mx + b \quad \text{Linear} \quad (3.1)$$

$$f(x) = \int_{-\infty}^{+\infty} e^{i\theta} d\theta \quad \text{Integral} \quad (3.2)$$

$$z = 2^k - \binom{k}{1} 2^{k-1} + \binom{k}{2} 2^{k-2} \quad \text{Binomial} \quad (3.3)$$

4 The Code Section

```

1 #
2 # 34567890123456789012345678901234567890123456789012345678901234567890
3 #
4 #!/usr/local/env python3
5 #
6 # m4tcolor.py
7 #
8 # How many mxm patterns are there on an nxn chess board n >= m?
9 #
10 def getPattern(i, m, n, x):
11     """Return the ith mxm pattern in a nxn board represented by x."""
12     assert n >= m, 'n < m'
13     pattern, gap, mask = 0, n - m, 2 ** m - 1
14     for j in range(m):
15         pattern += x >> i + j * gap & mask << j * m
16         #print('*', m, n, i, j, gap, i + j * gap,
17             # x >> i + j * gap, mask, mask << j * m, pattern)
18     return pattern

```

¹These correspond to the <dl>, , & HTML tags.

```

17
18 def getPatterns(m, n, x):
19     """Return the set of mxm patterns in a nxn board represented by x."""
20     assert n >= m, 'n < m'
21     patterns = set()
22     for i in range(n - m + 1):
23         for j in range(n - m + 1):
24             patterns.add(getPattern(i + j * n, m, n, x))
25     return patterns
26
27 def square(k, x):
28     """Return the unicode black or white square based on the kth bit of x."""
29     #return '\u25a0' if 1 <= k & x else '\u25a1'
30     return 'X' if 1 <= k & x else 'O'
31
32 def board(x, n):
33     """Return the string representation of x as an nxn board."""
34     return '\n'.join([ ''.join(
35         [ square(i + j * n, x) for i in range(n) ] for j in range(n) ])
36         for j in range(n) ])
37
38 tests = [ (2, 5), (4, 259), (5, 262149) ]
39 for m, n in tests:
40     f = 'm={ } n={ }'
41     print(f.format(m, n))
42     f += ' board={:0' + str(n * n) + 'b} {}=len({})'
43     for x in range(1000, 2 ** (n * n)):
44         patterns = getPatterns(m, n, x)
45         if len(patterns) == 2 ** (m * m):
46             print(f.format(m, n, x, len(patterns), patterns))
47             print(board(x, n))
48             break

```

Listing 1: m4taylor.py module

5 Conclusion

This last section provides a citation so that a bibliography is automatically included for (Bretscher, 2009) & (Eliot, 1871).

References

- Bretscher, O. (2009). *Linear algebra with applications* (4th). Pearson Prentice Hall. <http://isbn.nu/9780136009269>
- Eliot, G. (1871). *Middlemarch*. Penguin Books. <http://isbn.nu/9780141439549>